IIITB COMET FOUNDATION

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Gate -2011-IN -16 Question

QUESTION

For the Boolean expression $f = \bar{a}\bar{b}\bar{c} + \bar{a}b\bar{c} + a\bar{b}\bar{c} + abc + ab\bar{c}$, the minimized Product of Sum (PoS) expression is

(A) $f = (b + \bar{c}).(a + \bar{c})$

(B) $f = (\bar{b} + c).(\bar{a} + c)$

(C) $f = (\bar{b} + c).(a + \bar{c})$

(D) $f = \bar{c} + abc$

QUESTION ANALYSIS

Solution

Assuming the Boolean expression is

$$f = \bar{a}b\bar{c} + a\bar{b}\bar{c} + \bar{a}bc + abc + ab\bar{c}$$

Step 1: Convert to Minterms

$$\bar{a}b\bar{c} = m_2$$

$$\bar{a}bc = m_3$$

$$a\bar{b}\bar{c} = m_4$$

$$ab\bar{c} = m_6$$

$$abc = m_7$$

Therefore,

$$f = \Sigma m(2, 3, 4, 6, 7)$$

Step 2: Find the Zero Cells

The missing minterms are

$$m_0,\ m_1,\ m_5$$

Hence,

$$f = \Pi M(0, 1, 5)$$

Step 3: K-map Grouping of Zeros

AB\C	0	1
00	0	0
01	1	1
11	1	1
10	1	0

Group 1: m_0 **and** m_1 Common variables:

$$a = 0, \qquad b = 0$$

Maxterm:

$$(a + b)$$

Group 2: m_1 **and** m_5 Common variables:

$$b = 0, \qquad c = 1$$

Maxterm:

$$(b + \bar{c})$$

Step 4: Write the Minimized PoS Expression

$$\boxed{f = (a + b)(b + \bar{c})}$$

Verification

$$(a + b)(b + \bar{c})$$

is equal to 0 for

$$000, \ 001, \ 101$$

and equal to 1 for

$$010, \ 011, \ 100, \ 110, \ 111$$

which exactly matches the given function.

$$\boxed{\text{Minimized PoS} = (a + b)(b + \bar{c})}$$

Components Required

S.No	Component	Quantity
1	Pygmy FPGA Board	1
2	USB Cable	1
3	Computer/Laptop (Linux/Ubuntu/Termux)	1
4	FPGA Toolchain (QL SymbiFlow)	1
5	On-board LED	1

Hardware Connections

S.No	Connection	Description
1	FPGA Pin (LED Output) → Resistor	Current limiting resistor (220Ω–330Ω)
2	Resistor → LED Anode (+)	Connect resistor output to LED anode
3	LED Cathode (-) → GND	Connect LED cathode to Ground
4	FPGA GND → Circuit GND	Common ground reference

Execution Procedure

1. Create the Verilog source file `led`$_{t}ask.vandtheconstraintfile$

1. Compile the design using the QL SymbiFlow toolchain

```
ql_symbiflow -compile \
-d ql-eos-s3 \
-P pu64 \
-v led_task.v \
-t top \
-p led_task.pcf
```

2. After successful compilation, the bitstream file is generated inside the build directory.

```
build/top.bit
```

3. Rename the generated .bit file to a .bin file:

```
mv build/top.bit top.bin
```

4. Copy the generated .bin file to the computer where the FPGA programming environment has been established.

5. Open a terminal and navigate to the directory containing the generated binary file:

```
cd <path_to_binary_file>
```

6. Put the Pygmy FPGA board into Flash Mode and connect it to the computer using a USB cable.

7. Verify that the board is detected by the system:

   ```
   ls -l /dev/ttyACM0
   ```

   If the device is detected, information about the serial port will be displayed.

8. Program the FPGA board using the TinyFPGA programmer utility:

   ```
   python3 tinyfpga-programmer-gui.py \
   --port /dev/ttyACM0 \
   --m4app boolean_display_0.bin \
   --node n4
   ```

9. Wait until the programming process completes successfully.

10. Press the RESET button on the Pygmy FPGA board.

11. Observe the output on the hardware and verify the expected functionality.

Results and Observations

The generated Verilog design was successfully synthesized, implemented, and programmed onto the Pygmy FPGA board. After programming the FPGA and resetting the board, the LED outputs responded according to the logic combinations applied at the inputs A, B, and C.

Figure **??** shows the hardware setup and the corresponding LED output observed on the Pygmy FPGA board during testing.

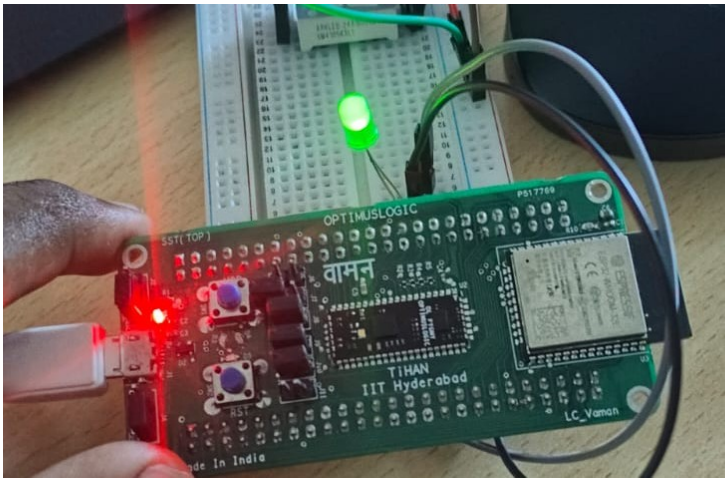

Figure 1: LED Output Observed on the Pygmy FPGA Board

The output was verified for different combinations of the input variables A, B, and C. The observed results matched the expected output obtained from the simplified Boolean expression

$$f = (a + b)(b + \bar{c})$$

The corresponding truth table is shown below.

A	B	C	LED Output (f)
0	0	0	0
0	0	1	0
0	1	0	1
0	1	1	1
1	0	0	1
1	0	1	0
1	1	0	1
1	1	1	1